STUMPRIDER · DESIGN DECISION RECORD

# StumpRider R7 judged on the bamboo local variant

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## DOCUMENT CONTROL

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| Document ID | SMR-DDR-004 | Version | 0.1 |
| Status | Draft | Date | 2026-10-04 |
| Author | Amish Chadha | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept | | |
| Repository | github.com/BoujeeEnjinia1701/stumprider | | |
| Source | docs/decisions/0004-r7-prototype-mass.md @ 507b36d | | |

## REVISION HISTORY

| Version | Date | Author | Change |
|---|---|---|---|
| 0.1 | 2026-10-04 | Amish Chadha | Amish's round-3 decision 2A recorded |

# 0004: R7 judged on the bamboo local variant

- **Date:** 2026-10-04
- **Status:** accepted. Decided by Amish on 2026-10-04: "For round 3, I agree with all your proposed recommendations". For StumpRider this accepts option A of open decision O5 in SMR-DEC-001 (portfolio item 2A).

## Context

The aluminium prototype carries 4.28 kg, 0.28 kg over R7's 4 kg (the 4.02 kg quoted earlier was 4.08 kg once lock pins and rivets were counted, and the hook head adds 0.21 kg). The bamboo local variant carries 3.89 kg.

## Options considered

A: keep the 32 x 2 aluminium pole and judge R7 on the bamboo local variant. B: a 32 x 1.6 aluminium tube (3.86 kg carried, buckling factor 1.84). C: restate R7 to 4.3 kg. Recommendation was A.

## Decision

R7 is judged on the bamboo local variant: 3.89 kg carried, met. The aluminium prototype's 4.28 kg is recorded, not a failure of R7, and is weighed at TRL 4. The pole tube is unchanged.

## Consequences

Requirements text, calculation note wording and registers only. No geometry, bill of materials or cost change. Value-engineering target: USD 2,000. Estimated cost of the constructable design: USD 299.50 (USD 1,700.50 under the target).